

Module 6: Sustainability

- **Case Study**
- Triple Bottom Line
- Life Cycle Analysis

The Four Functions of Management

- **Planning**
- Organising
- Leading
- **Controlling**

Case Study: Bougainville Copper

ENEL 301 2024

Washington Accord Graduate Requirements

Environment and Sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts.

The Bougainville Copper Project

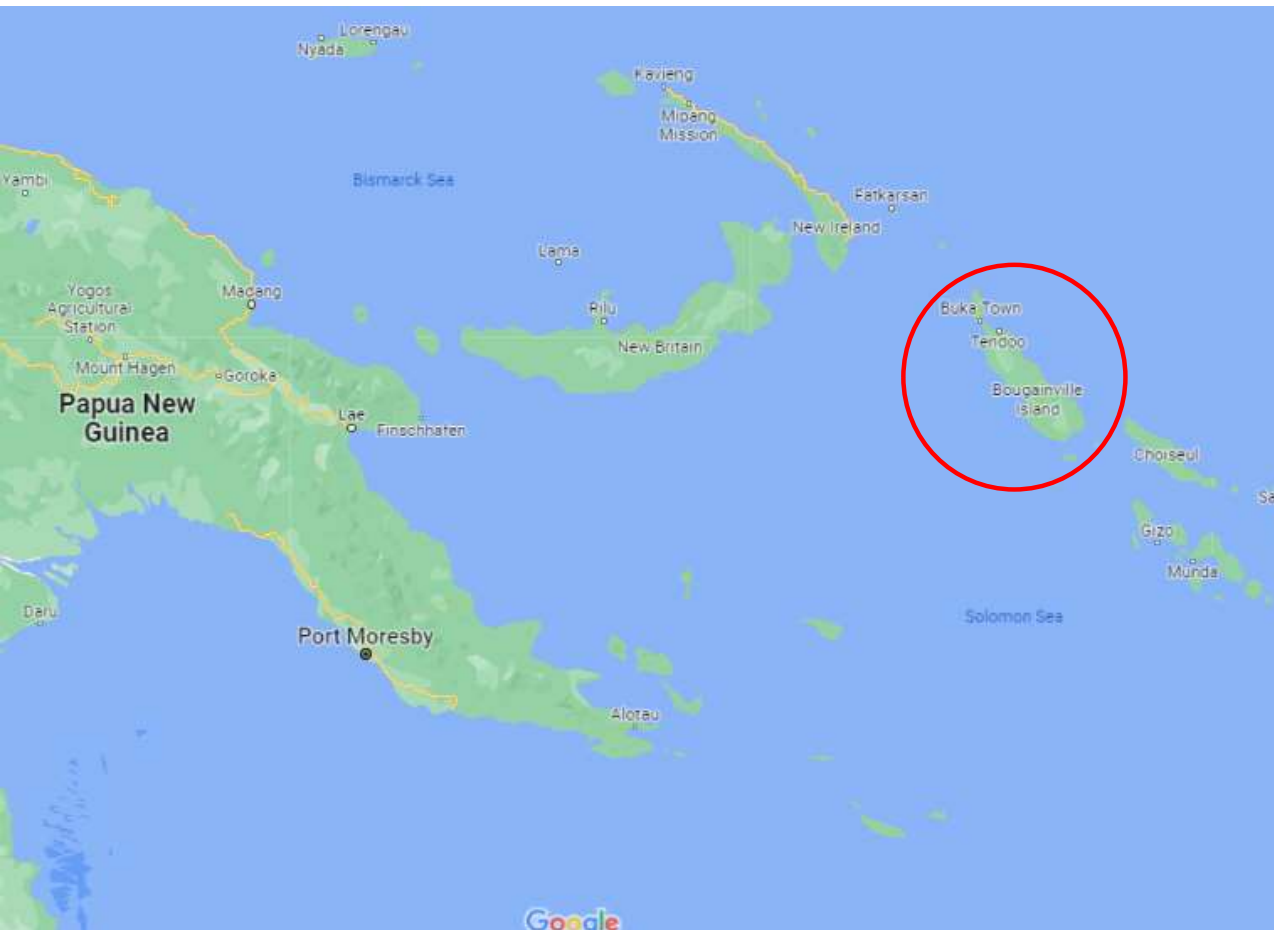
The Complete Project Life Cycle and Beyond

Environmentally / Socially / Economically

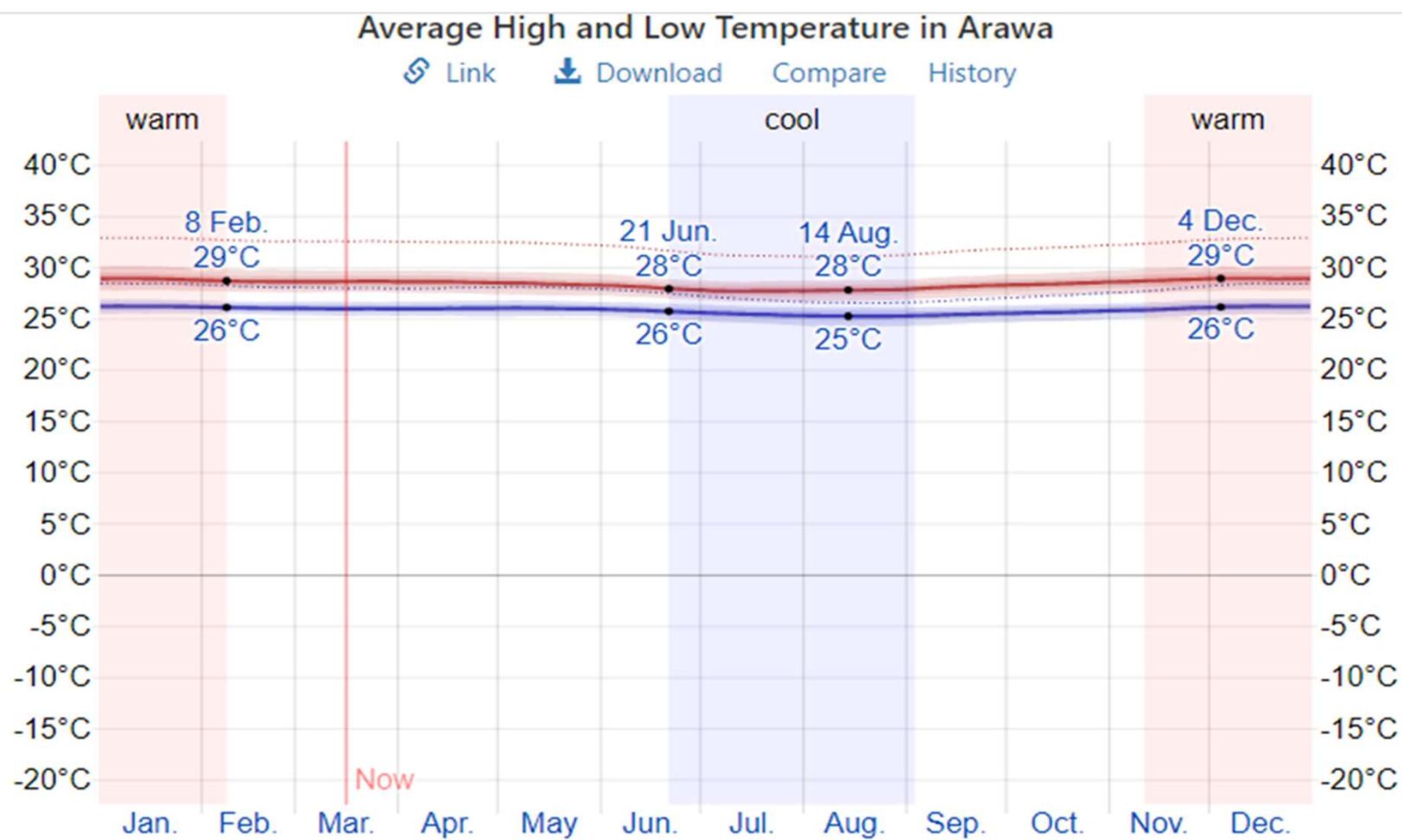


BOUGAINVILLE
COPPER LIMITED

Location

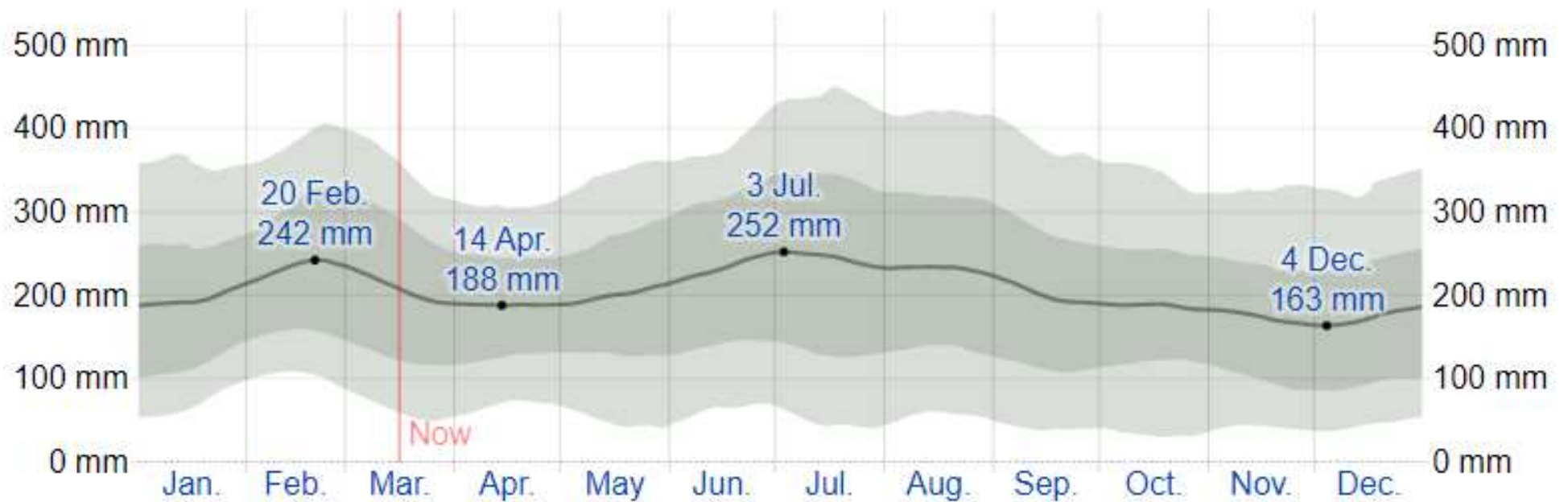






Source: <https://weatherspark.com>

Average Monthly Rainfall in Arawa

[Link](#)[Download](#)[Compare](#)[History](#)

Source: <https://weatherspark.com>



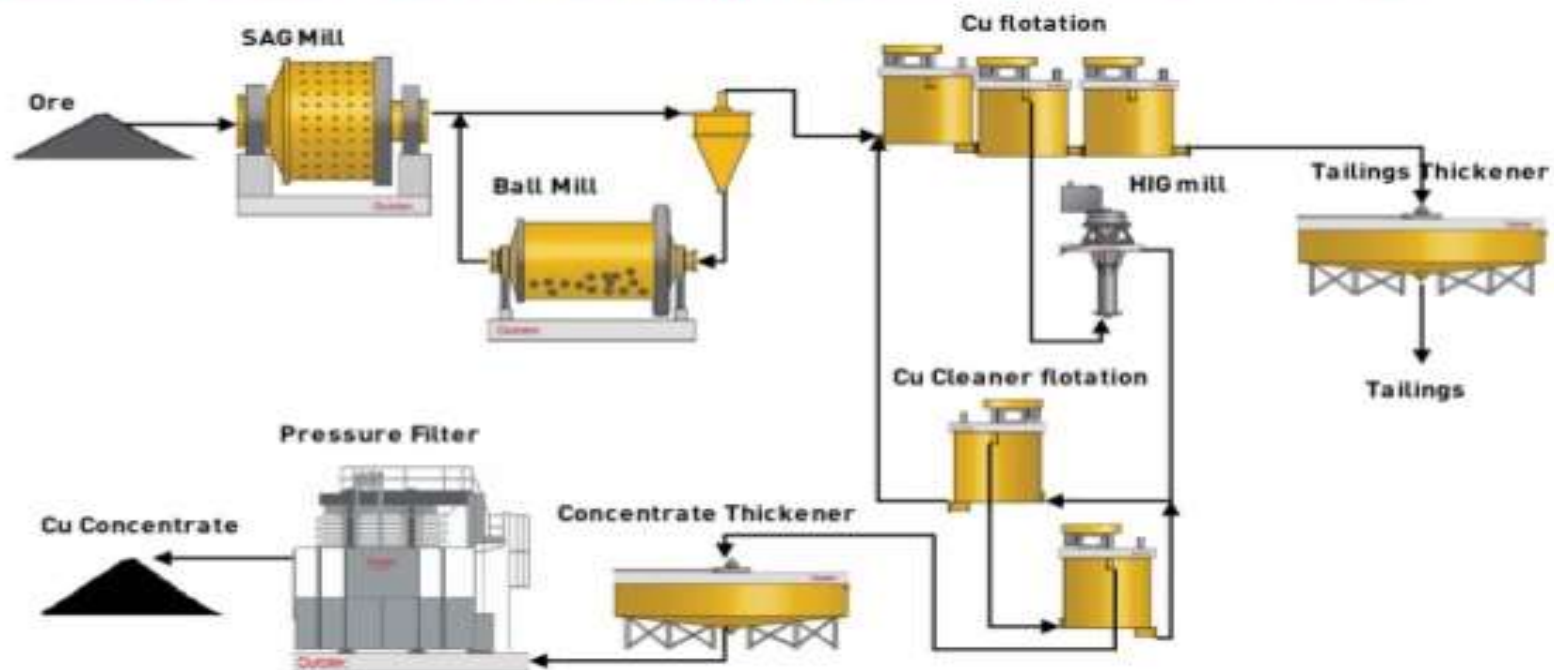






Porphyry Copper ore beneficiation

Example of the flowsheet for porphyry copper ore beneficiation



Outotec











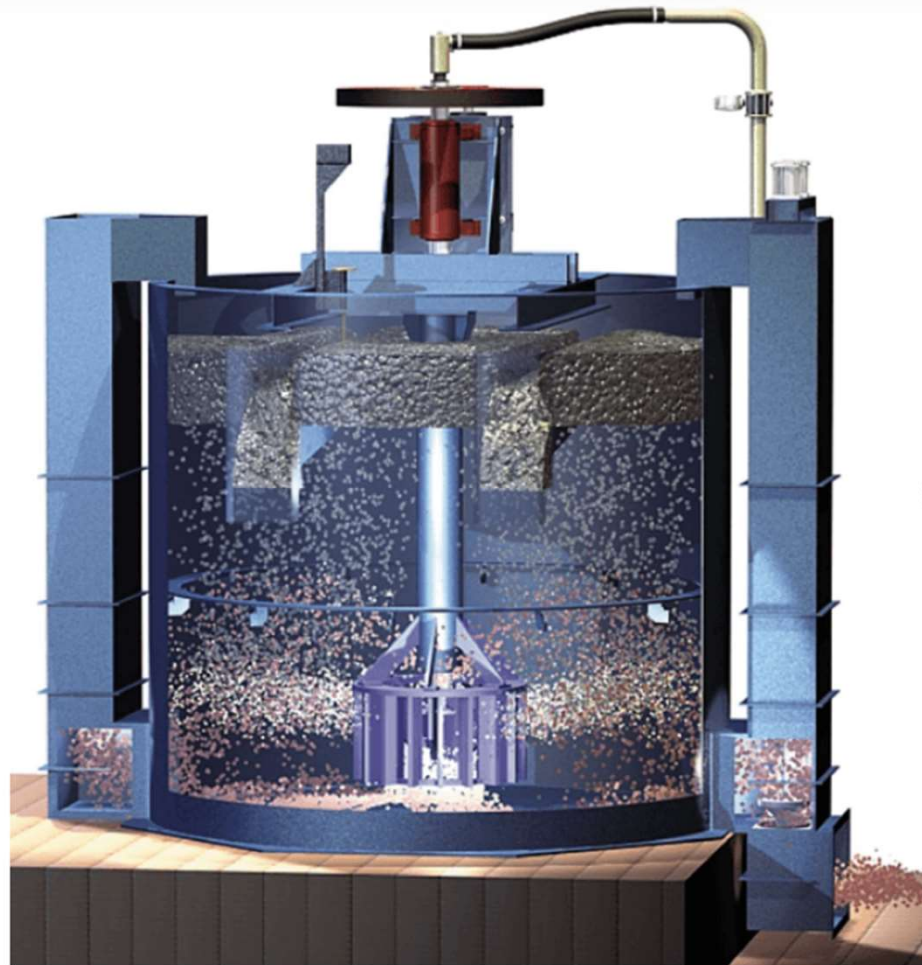
Motion Inside a Ball Mill ID x L
150mm X 110mm (6" x 4 1/3")
Using 10 mm (3/8") Steel Balls
@
Subcritical Speed of
85 RPM

ivw.de





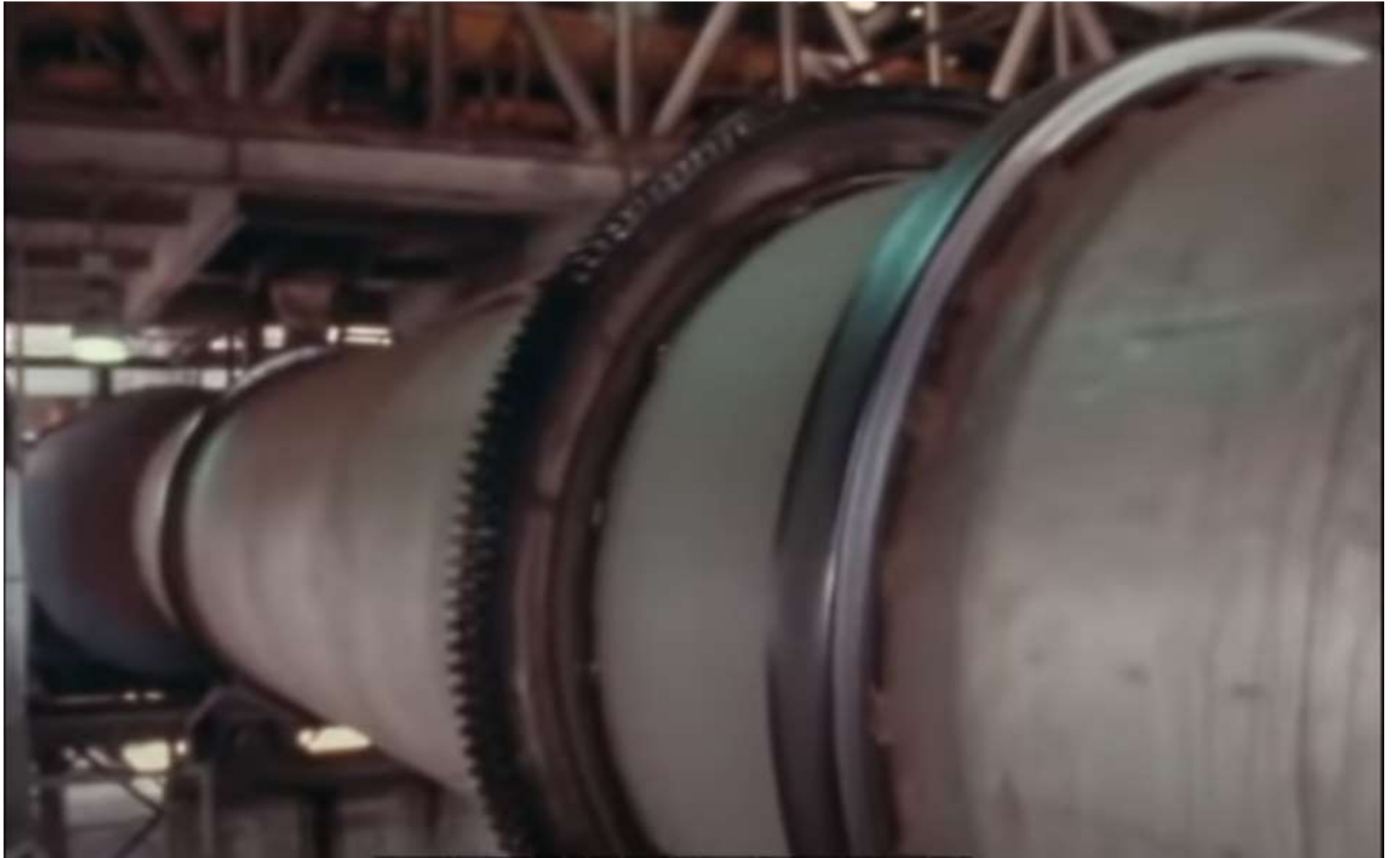
Flotation Cell













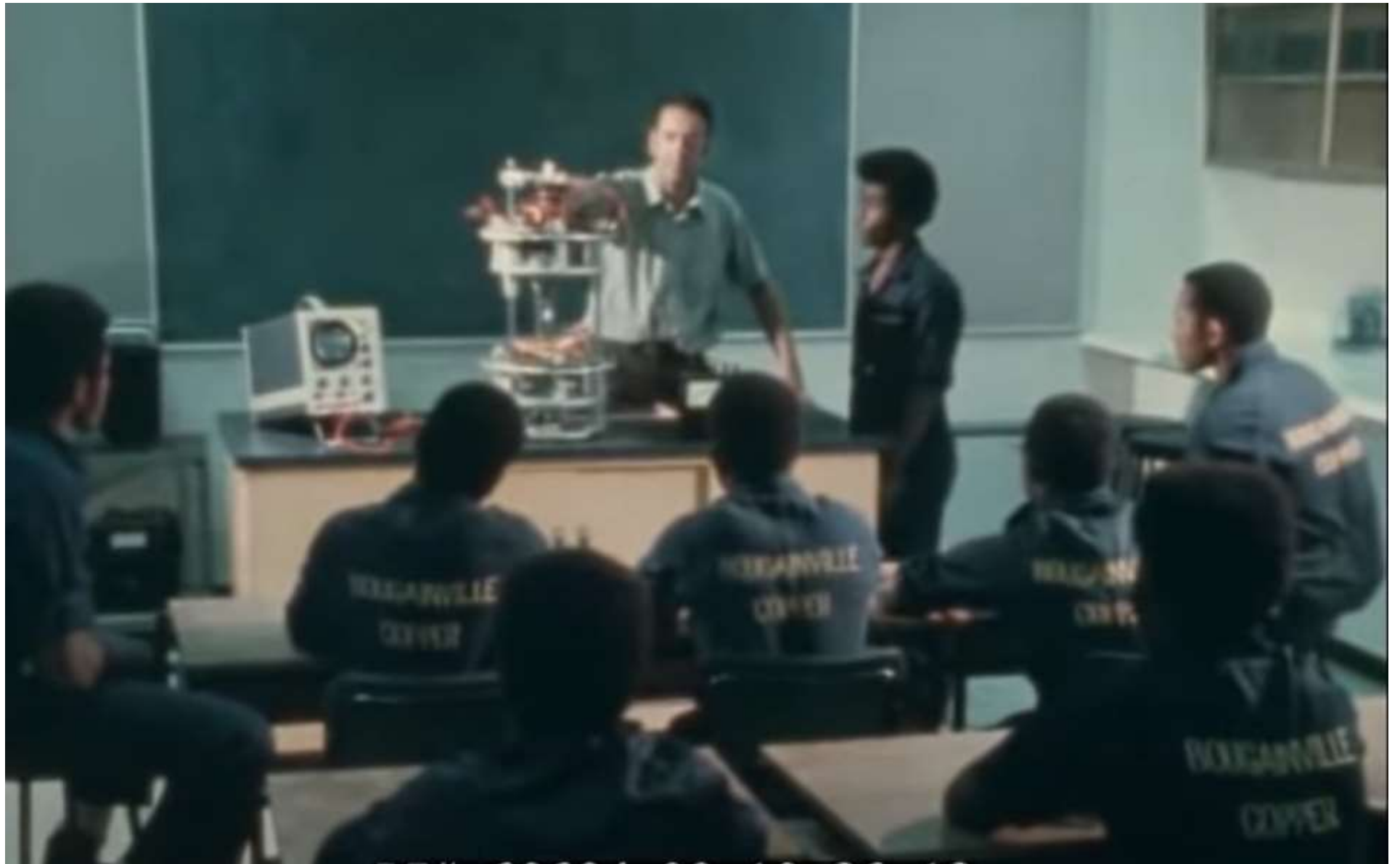




Social Impacts











Environmental Impacts





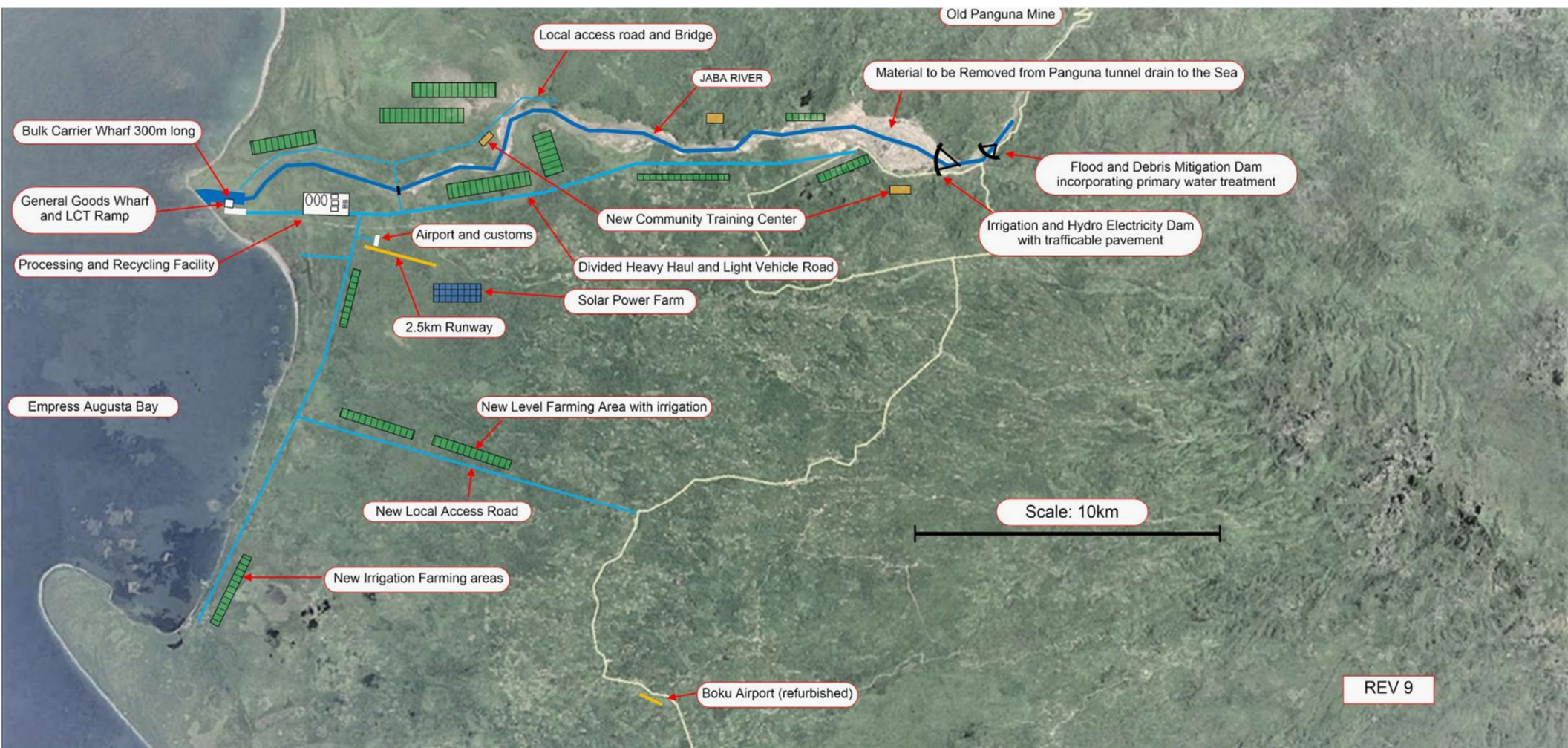
BUKA-TO-ARAWA HIGHWAY





The former Panguna Mine generated billions in revenue but this came at an enormous cost to the people and the environment

Jaba River Restoration – 3RE





Energy Transition Metals